



ENERGY MISCELLANEOUS — FULL MIND MAP

Topic-wise · Fully Explained · Hinglish · SSC / BPSC / BSSC / Railway

Topics in this chapter:

1. Site-Energy Match (Pavagada-Solar, Muppandal-Wind)
2. Rajasthan — Highest Solar Energy Potential
3. Power Plant Match (Chamera, Gwal Pahari, Kudankulam, Patratu)
4. Satara — Wind Energy (Chalkewadi Project)
5. Wind Power Potential — Tamil Nadu & Gujarat
6. Gujarat — #1 in Wind Power Generation
7. Non-Conventional Energy — Over 1% Contribution
8. Muppandal — Asia's Biggest Wind Project (150 MW)
9. Khambhat Coast — Largest Tidal Energy Area
10. Tidal Energy Potential Sites (Khambhat, Kutch, Sundarbans)
11. NHPC — Hydroelectric Power Corporation
12. Gujarat — #1 Installed Electricity Capacity
13. 1994 Per Capita Energy Consumption — 244 kg Oil Equivalent
14. Rampura (UP) — India's First Solar-Powered Village
15. SEB Financial Sickness — Multiple Causes
16. Manikaran — Geothermal Plant (Himachal Pradesh)
17. Jwalamukhi — Himachal Pradesh (Not J&K)
18. Biogas — Non-Commercial Energy Source
19. Renewable Sources — Water, Sun, Earth, Wind
20. Urja Ganga — Gas Pipeline (Eastern India, Not Iran)
21. Diu — First 100% Solar UT
22. Rajasthan — Leading Solar Energy Producer
23. Petroleum — Non-Renewable Fossil Fuel

| 1. Site-Energy Match (Pavagada-Solar, Muppandal-Wind)

- Match: Pavagada (Karnataka) — Solar, Tattapani (Chhattisgarh) — Geothermal, Muppandal (Tamil Nadu) — Wind, Kakrapar (Gujarat) — Atomic. [Q1]

| 2. Rajasthan — Highest Solar Energy Potential

- Ministry of New and Renewable Energy ke mutabik Rajasthan mein solar energy development ki sabse zyada potential hai (31 March 2024 tak). [Q2]

| 3. Power Plant Match (Chamera, Gwal Pahari, Kudankulam, Patratu)

- Match: Chamera — Hydro (Ravi river, Himachal Pradesh), Gwal Pahari — Solar (Gurugram, Haryana), Kudankulam — Atomic (Tamil Nadu), Patratu — Thermal (Ramgarh, Jharkhand). [Q3]

| 4. Satara — Wind Energy (Chalkewadi Project)

- Satara (Maharashtra) ka Chalkewadi wind energy project electricity generation ke liye famous hai. [Q4]

| 5. Wind Power Potential — Tamil Nadu & Gujarat

- Ministry of New and Renewable Energy ke mutabik **wind power development ki sabse zyada potential Gujarat, Rajasthan aur Tamil Nadu** mein hai — combined highest do options 'Tamil Nadu aur Gujarat' hain. [Q5]

6. Gujarat — #1 in Wind Power Generation

- 31 March 2024 tak wind power generation capacity mein **Gujarat pehle number par hai, Tamil Nadu dusre number par**. [Q6]

7. Non-Conventional Energy — Over 1% Contribution

- Sahi statement: **Industry India ka major energy-consuming sector hai — 'non-conventional sources 1% se kam contribute karte hain' GALAT hai (ab 1% se zyada hai)**. [Q7]

8. Muppandal — Asia's Biggest Wind Project (150 MW)

- Jab question pucha gaya tha, Asia ka sabse bada wind energy project **Muppandal village (150 MW), Tamil Nadu** mein tha. [Q8]

9. Khambhat Coast — Largest Tidal Energy Area

- India ka **sabse bada tidal energy producing area Khambhat Coast (Gulf of Cambay)** hai. [Q9]

10. Tidal Energy Potential Sites (Khambhat, Kutch, Sundarbans)

- Tidal energy ke 3 most-potential locations: **Gulf of Cambay/Khambhat, Gulf of Kutch (west coast), Ganges Delta/Sundarbans (east coast)** — Bhavnagar, Khambhat ki khaadi mein hi sthit hai. [Q10, Q11]

11. NHPC — Hydroelectric Power Corporation

- NHPC (National Hydroelectric Power Corporation, 1975 incorporate hui) **electricity generation se related hai** — baad mein geothermal, tidal, wind sources bhi include kiye. [Q12]

12. Gujarat — #1 Installed Electricity Capacity

- 31 Dec 2024 tak **gross electricity installed capacity** mein Gujarat pehle number par hai. [Q13]

13. 1994 Per Capita Energy Consumption — 244 kg Oil Equivalent

- 1994 mein India ki per capita energy consumption **244 kg oil equivalent** thi — world standard (1471 kg) se bahut kam. [Q14]

14. Rampura (UP) — India's First Solar-Powered Village

- Rampura, **Uttar Pradesh** mein hai — apna solar power plant lagane wala India ka pehla gaon bana. [Q15]

15. SEB Financial Sickness — Multiple Causes

- State Electricity Boards (SEBs) ki financial sickness ke sab 4 reasons sahi hain: **agriculture/domestic consumers ko cost se kam par bijli bechna, high transmission-distribution losses, commercial autonomy ki kami, social subsidy policies**. [Q16]

16. Manikaran — Geothermal Plant (Himachal Pradesh)

- Manikaran geothermal power plant **Parvati Valley, Kullu district, Himachal Pradesh** mein hai. [Q17]

17. Jwalamukhi — Himachal Pradesh (Not J&K)

- Manikaran-HP, Anthoni-MP, Tapta Pani-Odisha sahi hain — '**Jwalamukhi-Jammu&Kashmir**' **GALAT** hai (Jwalamukhi asal mein Himachal Pradesh ke Kangra district mein hai). [Q18]

 **EXAM TRAP — Jwalamukhi — Himachal Pradesh mein hai, J&K mein NAHI**

18. Biogas — Non-Commercial Energy Source

- Commercial energy sources: **Petroleum, Atomic Energy, Natural Gas, Coal** — **Biogas non-commercial (renewable) source** hai. [Q19]

19. Renewable Sources — Water, Sun, Earth, Wind

- Sab renewable sources hain: **Water, Sun (solar), Earth (geothermal), Wind** — koi bhi option akela sahi nahi tha, isliye 'None of the above' sahi answer hai. [Q20]

20. Urja Ganga — Gas Pipeline (Eastern India, Not Iran)

- Urja Ganga project **gas pipeline** hai, **October 2016** mein launch hua (PM Modi ne Varanasi mein foundation rakha) — Eastern India ko connect karta hai, Iran-India se **NAHI**. [Q21]

21. Diu — First 100% Solar UT

- Diu **India ka pehla Union Territory** hai jo **100% solar energy** par chalta hai (~40 sq km area) — 26 Jan 2020 se Daman-Diu aur Dadra-Nagar Haveli merge hue. [Q22]

22. Rajasthan — Leading Solar Energy Producer

- 31 March 2024 tak **solar power production** mein Rajasthan **sabse aage** hai. [Q23]

23. Petroleum — Non-Renewable Fossil Fuel

- Petroleum (crude oil) ek **non-renewable fossil fuel** hai — coal aur natural gas ki tarah ancient marine organisms se bana hai. [Q24]